FOOLED AGAIN

A few little words is all it takes to turn your rational brain to jelly. Fortunately, **Michael Brooks** has a perfectly logical explanation

ON 1 SEPTEMBER 1983, Korean Airlines flight 007—bound for Seoul—strayed inadvertently into Soviet airspace. The reaction from the Soviet authorities was swift. Two fighter planes were scrambled to intercept it and one fired an air-to-air missile that ripped through the airliner's fuselage and sent it plummeting into the Sea of Okhotsk. Two hundred and sixty-nine people were killed.

Despite the international outcry, the Soviets remained unrepentant. An unidentified plane had strayed into their airspace at night. They believed they had every right to "stop the flight". But in spite of the decades of inquests and analysis, nobody has been able to resolve why the airliner strayed so far from its designated course. Why didn't the pilots notice their error and correct it?

The flight plan told the pilots what to expect. They knew that, if the plane is on course, then the radar would show only water. For at least 25 minutes the pilots could see that the radar was showing the land mass of the Kamchatka Peninsula. But the crew did not draw what appears to be the obvious conclusion. They maintained their heading and, minutes later, the Soviet fighter plane struck.

It seems an extraordinary error. But the same breakdown of logic might also have been behind the meltdown at Chernobyl. Research published earlier this year claims that these failures in reasoning are a common occurrence, arising whenever we are faced with scenarios that include "falsity"—things that may not be true. For flight 007, the results were catastrophic. But according to Philip Johnson-Laird, a psychology professor at Princeton University, we also encounter the same logical meltdown in events both serious and trivial throughout our lives.

You might have experienced this logical breakdown while hiking or driving with the aid of a map. If you are on course, then the landscape you see corresponds to the features the map tells you to expect. But if you find yourself off-course, working out your location—and the way back to the right road—gets much more

Illustrations: Chris Draper

difficult. You have to deal with false situations: if you had been on the right track you would have seen a gate leading into a wood, for example. But you didn't, and attempting to compare what you didn't see with what you should have seen leads you easily into confusion. Eventually, you give up on the logical solution to your problem and head onwards. When you do see something that relates to the map working out your whereabouts becomes trivial. That's because it's easier to deal with a true scenario than a false one.

For more than a decade, Johnson-Laird has been fighting against the received wisdom among researchers who investigate human powers of reasoning. The widely accepted model for the way we reason is known as "formal rules". According to this hypothesis we use rational, logical rules when we think things through.

Johnson-Laird concedes that we are certainly capable of this kind of thought: with care, we can slavishly follow logical rules of deduction. However, he believes we don't usually think that way. Yes, we can be rational if we really put our minds to it, but we usually employ shortcuts that save us a lot of time and effort. These shortcuts, he says, are our everyday mode of thinking. Unfortunately, he adds, they can also lead us into making foolish mistakes.

Instead of pursuing a set of rules and following through every piece of information to its logical conclusion, Johnson-Laird believes we construct "mental models"—imaginary sketches of the possibilities of a situation—and work from these. To test his hypothesis, Johnson-Laird constructs deceptively innocent logic puzzles like the one below.

Only one of the following statements about a particular hand of cards is true:

- There is a king in the hand, or an ace, or both.
- There is a queen in the hand, or an ace, or both.
- There is a jack in the hand, or a ten, or both.
- Is it possible that there is an ace in the hand?

On first examination the answer seems obvious: yes. But ask a formally programmed computer the same question and it will tell you the opposite. If there is an ace, then the first two statements are true. But the puzzle explicitly states that only one statement is true. So an ace is not possible.

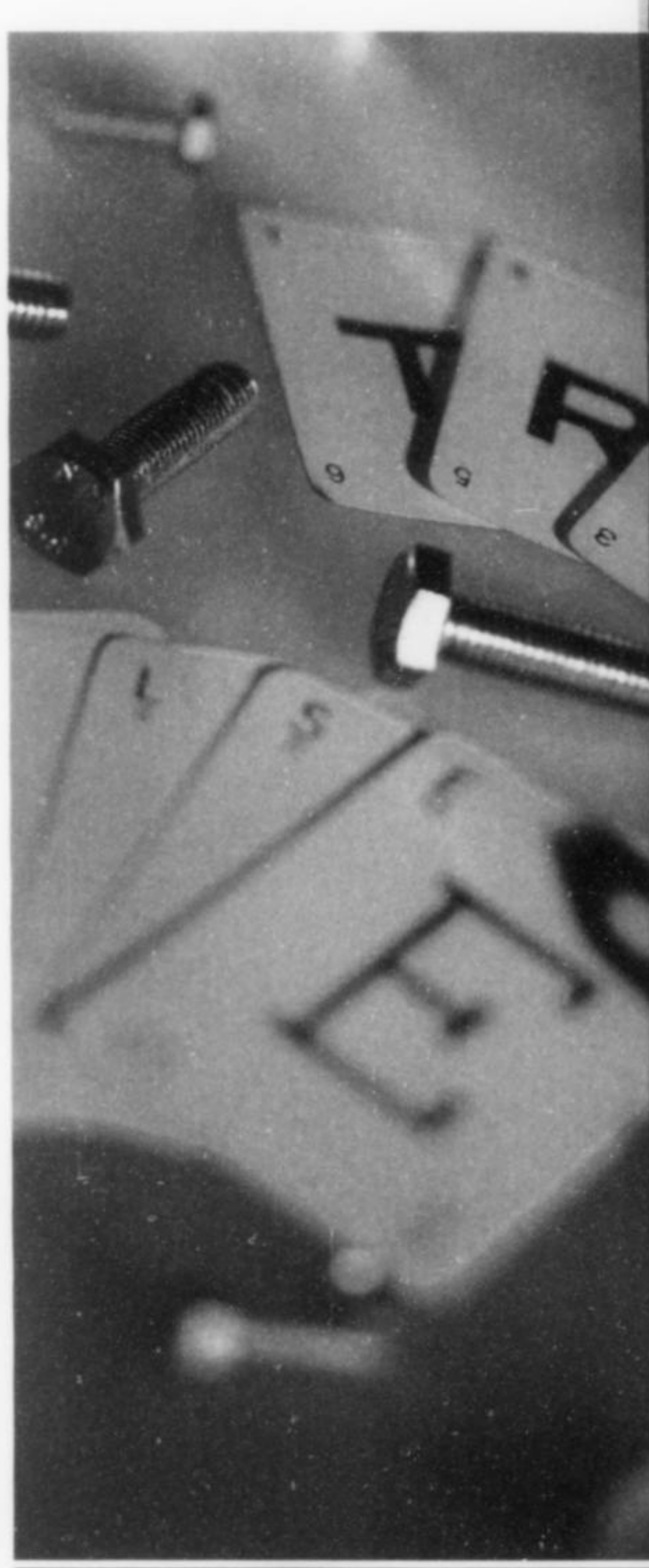
Johnson-Laird tested Princeton University students with this problem, and 99 per cent of them got it wrong. The reason for that extraordinary degree of error, he says, is that there is limited space in what researchers call "working memory": the low-capacity, short-term memory that supports language, arithmetic and reasoning. When we draw our mental models of a situation our working memory runs out of space very quickly. So, to save time, space and effort, we leave vital information off the "drawings". The pictures are all there, but the labels—like "this picture is only true if the other picture is false"—can go missing.

The first casualty of a full memory is anything that's not true, says Johnson-Laird. People can cope with the potential falsity of single-clause sentences, such as "Pat loves Val". If someone says that's untrue, it's clear what they mean. "But they are not so hot with the potential falsity of 'John is tall and Mary is short'," he says. If we are told that this statement is false there are suddenly a lot of options to consider. Does it mean that John is short, or that Mary is tall, or that neither is tall or short, or that we can't draw any conclusion about their heights? When anything but the simplest situation involves falsity the number of possible scenarios quickly becomes too great to hold in working memory. So, Johnson-Laird claims, we ditch the falsity and hope for the best.

Take the playing card puzzle above. In that case, the mental model will be:

• king	ace	king and ace
• queen	ace	queen and ace
• jack	ten	jack and ten

Using this model, we assume that it is definitely possible to have an ace: there it is, looming large in two of the lines. But what doesn't make it into the mental model is the fact that one—and only one—of the lines is true, so the other two are false. "When people think aloud about this problem, they clearly realise that one assertion is true and the others are false," says Johnson-Laird. But that doesn't make it into their mental model because it involves setting up a whole new set of models. When you consider the case in which it is true that there is a king or an ace or both, you must also work out what follows from the falsity of the other assertions. All these possibilities have to be modelled, and suddenly there's just too much information to handle. So, instead, we stick with our "instinctive" answer—which is wrong. Here the consequences are not particularly grave. But when the



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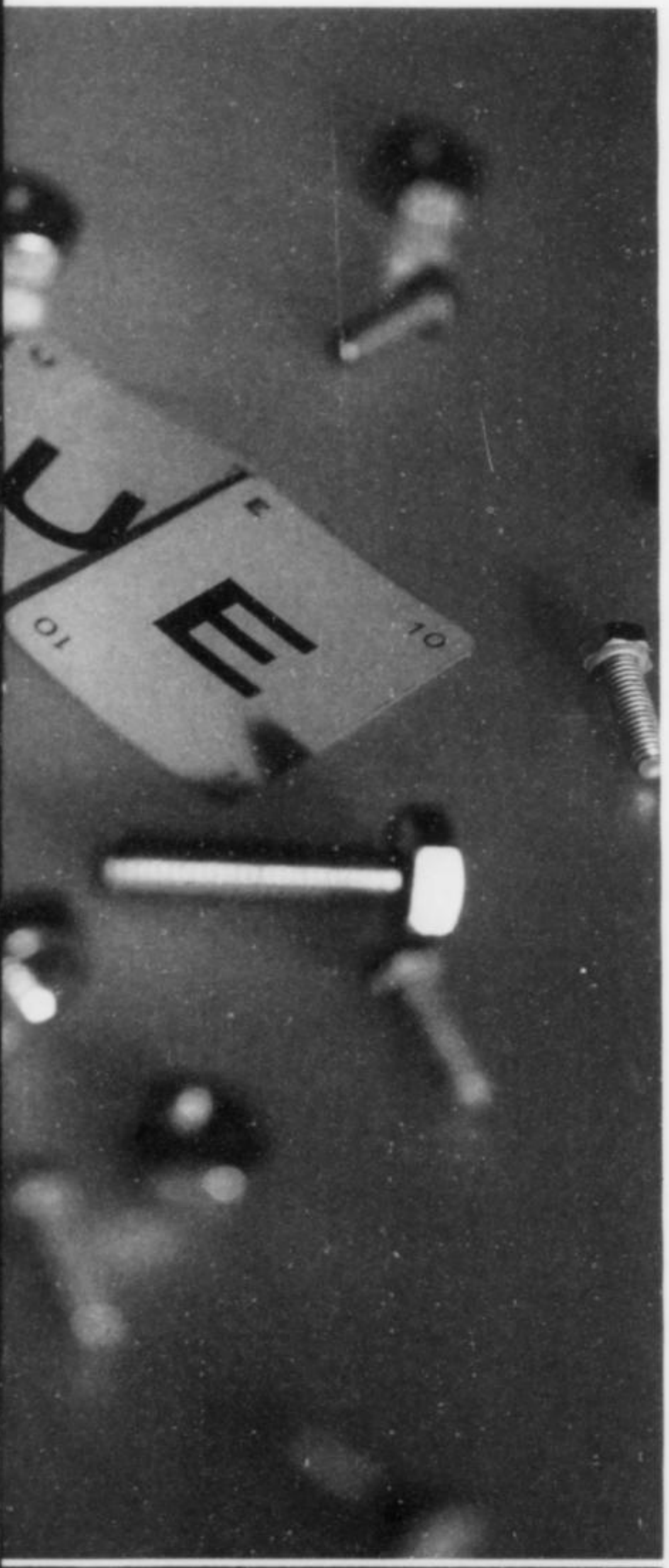

Meltdown: debris from an abandoned building near Chernobyl. The nuclear accident could have been a consequence of faulty logic

Korean pilots took the same shortcut, it cost 269 lives.

Not persuaded? Then look at the puzzles Johnson-Laird devised to demonstrate the power of the illusions the mind can create. The question in each test is, can both statements be true at the same time?

Test 1
There is a pin and/or a bolt on the table, or else there is a bolt and a nail on the table.
There is a bolt and a nail on the table

Test 2
There is a pin and/or a bolt on the table, or else there is a bolt and a nail on the table.
There is a pin and a bolt on the table

Test 3
There is a pin and/or a bolt on the table, or else there is a bolt and a nail on the table.
There is a nail but no bolt on the table

Test 4
There is a pin and/or a bolt on the table, or else there is a bolt and a nail on the table.
There is no pin and no bolt on the table

In these four tests there is an inconsistent pair of statements, an illusion of consistency, an illusion of inconsistency and a consistent pair. They are not laid out in that order, however, and—unless you can think like a computer—it is not at all obvious which is which.

In April this year, Johnson-Laird reported his results from performing these tests on more than 500 students (*Science*, vol 288, p 531). Johnson-Laird found that all but one of them were fooled by the illusions in precisely the way the mental model theory predicted they would be.

Test 1 is an illusion of consistency. When you work this problem out, Johnson-Laird believes, you make a "mental model" of the four possibilities from the first statement:

pin
bolt
pin bolt
bolt nail

Here you have four possibilities, and the fourth is consistent with the second statement that there is a bolt and a nail on the table. So it looks as though the statements are consistent. The illusion lies in a couple of little words that don't make it onto the mental model: "or else". If the first half of the statement is true, then the "or else" means the second half is false, and vice-versa. But we discard this complicated falsity, and process the problem without it. We assume everything on the model is possible and fall for the illusion.

Picture it like this. If we took the "or else" into account, this would give the following mental model:

pin
bolt
pin bolt
bolt nail

where everything in black is only true if everything in red is false, and vice-versa.

If the first half of the statement (everything in black) is true, then the "or else" means the second half is false and there cannot be a bolt and a nail on the table. So the second statement ("there is a bolt and a nail on the table") is then inconsistent with the first. And if everything in the first half is false, there is neither a pin nor a bolt on the table. Therefore there can't be a bolt and a nail.

Test 2 replaces the second statement with "there is a pin and a bolt". Now the two statements become consistent. You'll find you get that right easily, because the "or else" doesn't need to be processed.

Test 3 is an illusion of inconsistency. If the second half (in red in the mental model above) is false, there can be a nail without a bolt. That fits perfectly well with a true first half too. So there's no inconsistency between the two statements.

Test 4, by the way, is the inconsistent pair of statements: whichever half of the first part is true, there is always either a pin or a bolt on the table.

How does Johnson-Laird know that his subjects made their mistakes because they failed to grapple with the notion of falsity? What if their working memory were simply filling up randomly? Johnson-Laird's confidence comes from the predictable nature of the errors people make. By invoking the problem with falsity, he was able to anticipate exactly what mistakes the students would make on each of the tests. "If human beings were intrinsically rational, they should make only sporadic errors similar to slips of the tongue," he says.

So maybe people are using invalid logical rules? No, he says, because that would make people intrinsically irrational—and we're not: we understand our mistakes when they are explained. The only answer, Johnson-Laird says, is that we lose the "or else". That gets rid of the complex

'Such deliberate counterfactual thinking might also be at the root of creativity. Imagination and daydreaming involve creating partially false situations'

falsity and makes our lives easier.

We may have evolved to discard the falsity, since it saves a lot of mental effort. Often we only need to think about those situations that are happening in reality, and are therefore true. But when falsity does matter—as it did for the Korean pilots—that evolutionary adaptation can turn on us spectacularly.

Johnson-Laird believes the pilots' mental models of the situation with the radar were lacking vital information. They probably looked something like this:

On course Water
Land

There is nothing in this model to depict the possibility that "the plane is on course" is false. According to the report of the International Civil Aviation Organization, the pilots on the Korean airliner may have been suffering from fatigue, a situation in which they would be conserving every ounce of mental energy. With a conditional "if...then" statement, we are perfectly aware that the clause following "if" could be untrue. But we don't always make the effort to flesh that possibility out. Johnson-Laird believes we often leave it as an unexplored path in the back of our minds.

So when the land contradicted the possibility of the plane being on course, the pilots had nothing but that unexplored path before them. "It is as though they think to themselves, 'there are other possibilities, but I won't worry about them now'," says Johnson-Laird.

In laboratory tests that mimic this scenario, Johnson-Laird says, many people also draw no conclusion when confronted with this kind of falsity. The operators conducting the experiment that led to disaster at Chernobyl may have been in exactly the same position. According to Zhores Medvedev's 1992 book on the meltdown, *The Legacy of Chernobyl*, they had the following pieces of information:

"If the experiment is safe to continue, then the turbines must be rotating fast enough to generate emergency electricity."
"The turbines are not rotating fast enough to generate emergency electricity."

From this, they should have recognised that the first clause could be false, and admitted into their mental model the possibility that the experiment might no longer be safe. But they didn't. "They went ahead with the experiment—with tragic consequences," says Johnson-Laird.

The mental models theory has yet to convince everyone who researches reasoning and deduction. Ira Noveck of the Claude Bernard University in Lyon, for example, believes its conclusions may result simply from people's difficulty interpreting the language of the puzzles. At least with the mental rules model of reasoning, he says, it is easier to work clearly through the mechanisms behind reasoning. Johnson-Laird insists he has conducted numerous variations on the basic tests using different wording, and the results come out just the same. And when people are asked to describe their thinking as they work through the puzzles, this also suggests that they understand exactly what they're being asked to do.

Ruth Byrne, a mental models researcher at Dublin University who is collaborating with Johnson-Laird, says the illusions provide a powerful experimental proof for the mental models theory. "It's compelling evidence," she says. "It can't be explained by any other theory of reasoning."

Byrne is investigating how mental models are involved with emotions such as regret and guilt. These emotions require a deliberate effort to model falsity: they rely on considering possible alternatives to the real-life consequences of events. "You couldn't explain an emotion like regret unless you were keeping in mind the way a situation turned out and comparing this with an alternative where it could have turned out differently," Byrne says.

Such forced, deliberate "counterfactual" thinking might also be the root of creativity, she adds. Byrne believes that imagination and daydreaming involve creating these partially false situations and working through the outcomes of their models. But that won't help you pen a bestselling novel, because most of us tweak our mental models in exactly the same ways. "There is really remarkable regularity in the counterfactual alternatives that people generate," Byrne says. "What we seem to find more difficult is creating counterfactual alternatives that are unlike the ones that everyone else creates."

Find this elusive ability and you've struck gold. But beware: by now you should have realised that dealing with falsity comes at a price. Guilt, regret and writing a novel are renowned as some of humanity's favourite instruments of self-torture. At least we understand now why it hurts. □